

- 4 (a) (i) Using Fleming's left hand rule, the magnetic force acting on the electron is always perpendicular to the direction of its linear velocity. Hence the direction of the electron's velocity changes, but its speed remains constant.

Since magnetic flux density  $B$ , charge of electron  $e$  and speed of electron  $v$  are constant, the magnitude of the magnetic force  $Bev$  remains constant.

This magnetic force provides the centripetal force and hence the electron will be in uniform circular motion.

- (ii) The magnetic force on the electron provides the centripetal force.

$$\begin{aligned} Bqv &= mr\omega^2 \\ Bq(r\omega) &= mr\omega^2 \\ Bq &= m\omega \\ Bq &= m(2\pi f) \\ f &= \frac{Be}{2\pi m_e} \\ &= \frac{(2.8 \times 10^{-5})(1.60 \times 10^{-19})}{2\pi(9.11 \times 10^{-31})} \\ &= 7.8267 \times 10^5 = 7.83 \times 10^5 \text{ Hz} \end{aligned}$$

- (b) (i)  $Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{Bq} \Rightarrow r \propto v$  (since  $B$ ,  $m$  and  $q$  are constant)

The speed of the second electron is  $\sqrt{2}$  times the speed of the first electron. Since  $r \propto v$ , the radius of its circular path is  $\sqrt{2}$  times the radius of the circular path of the first electron.

(ii) 
$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \left( \frac{mv}{Bq} \right) = \frac{2\pi m}{Bq}$$

The period of revolution is independent of the speed of the electron. The period of revolution for both electrons is the same.

- (iii) The magnetic force is always perpendicular to the direction of the linear velocity / displacement of each electron. There is no displacement in the direction of the magnetic force and the work done by the magnetic force on each electron is always zero.

- (c) The electric and magnetic fields are pointing perpendicularly into the paper. The electron experiences a constant electric force directed perpendicularly out of the paper while it undergoes uniform circular motion.

The electron will accelerate out of the paper in a helical path with increasing pitch.

**Raffles Institution**  
**Year 5-6 Physics Department**